Critical review of the GRBV analysis pipeline

Simulated journal-reviewer critique of the GRBV / Vitis vinifera cv. Merlot manuscript, benchmarked against published plant RNA-seq papers in target journals (Nature Plants, Plant Cell, New Phytologist, PNAS).

Severity key. 🔴 Critical (must fix before submission) 🟠 Major (should fix) 🟢 Minor (nice to have)

Author: Prem Pratap Singh • **Date:** 2026-04-20 • See companion *INTEGRATED_ANALYSIS*, *SENSITIVITY_REPORT* and *HSFA2_VIGS_EXPERIMENT_PLAN* documents.

Critical review of the GRBV analysis pipeline

Simulated journal-reviewer critique. Every issue below is something a peer reviewer at Nature Plants / Plant Cell / New Phytologist / PNAS is likely to raise. Issues are graded by severity; remediation steps specify what to rerun and against what published benchmark. ***Read this before the first manuscript submission.***

Severity key

• ■ **Critical** — must fix before submission. A reviewer will reject or return for major revision.

• ■ **Major** — should fix before submission or have a defensible answer for rebuttal.

• ■ **Minor** — nice to address; strengthens the paper but unlikely to tank review.

1. Differential expression methodology ■

What the pipeline does

• `analysis.py` and several downstream scripts run **Welch's *t*-test on `log2(CPM+1)`** with BH FDR and `|log■FC| ≥ 1`.

Field standard (what reviewers will expect)

Bulk RNA-seq DE in plant-virology papers published in Nature Plants, Plant Cell, New Phytologist, and Plant Journal in the last 5 years almost universally uses one of:

1. **DESeq2** (Love et al. 2014) — negative binomial GLM on raw counts with size-factor normalisation and `lfcShrink` (apeglm / ashr).
2. **edgeR** — quasi-likelihood F-test (`glmQLFit` / `glmQLFTest`) with TMM normalisation on raw counts.
3. **limma-voom** — mean-variance modelling on log-CPM, then empirical Bayes moderation with `eBayes`.

Welch's *t*-test on log-CPM **does not model the mean-variance trend of count data**, so it under-powers low-count genes and over-rejects high- count ones. It also does not shrink variance across genes, meaning the per-gene SD is noisy with n ≈ 30–60 per group. Reviewers will ask why a bulk-RNA-seq study with hundreds of samples used a generic 2-group test rather than the field-standard negative-binomial / voom approach.

Remediation

• **Rerun DE for Q1–Q6** with DESeq2 *and* limma-voom in parallel. Report overlap with the Welch-t results as a robustness check in the supplement.

• Expose the DE design matrix: `~ Sampling_Season + Maturity_stage + Infection_Group + Infection_Group:Maturity_stage` for Q2/Q4. Current analysis treats each pairwise comparison in isolation, losing power.

• Re-calculate the **908 universal DEGs** after the above.

Expected outcome: core findings (sHSP suppression, immune silence, sugar rewiring, biphasic trajectory) will survive because the effect sizes are large. But specific gene lists will shift. This is a 1–2 day compute + write job.

2. Experimental-design reporting ■

Issues

1. **No power calculation in the methods.** With 63 RB− controls vs 264 RB+ infected, the group imbalance (4:1) is unusual. Reviewers will ask whether the control sample size was pre-specified.
2. **No allocation table.** How many samples per (season × stage ×

infection-cohort) cell? A ≥ 12-row allocation table is standard.

3. **Repeated-measures structure not modelled.** Several samples come

from the same vine across stages / seasons. The current DE model treats every sample as independent. For the Q2 stage-based comparisons and Q6 trajectory analysis, this is a genuine pseudoreplication concern.

4. **Single vineyard, single cultivar** (Merlot) — a field limitation,

not a pipeline issue, but should be explicitly flagged in the Discussion with "limits generalisability" language.

Remediation

- Add allocation table to supplement.
- Refit DE with `~ condition + (1 | vine_id)` or equivalent in `lmerSeq`

/ limma-voom with `duplicateCorrelation`. If vine identity isn't available, document why and use block bootstrapping in a sensitivity check.

- Explicit "limits of generalisation" paragraph in the Discussion.

3. Read-level QC and alignment not documented in the repo ■

Issue

The Methods section a reviewer would expect covers:

- Read QC (FastQC, MultiQC)
- Adapter trimming (Trim Galore / fastp)
- Alignment / quantification (STAR + featureCounts, Salmon / Kallisto,

or HISAT2 + StringTie)

- rRNA / chloroplast / GRBV decontamination
- Alignment rate, assigned-reads rate, mean read count per sample
- Multi-QC plots for the cohort

None of this is in the repository or documented. Reviewers will demand it. Even if the work was outsourced (commercial quant), the pipeline version, reference genome + annotation, and alignment statistics must be reported.

Remediation

- Recover or reconstruct the QC/alignment metadata. Generate a

MultiQC report for the 327 libraries.

- State the reference genome (NCBI GCF_030704535.1 is what the

extended pipeline uses — confirm this matches the original quant) and annotation version explicitly.

- Deposit raw FASTQ or at least counts + metadata to GEO / SRA.

4. Causal mediation analysis as published ■

Issue

The main paper's `causal_mediation_analysis.py` filters to **top-500 variable genes** before the mediation regression (line 253: `nlargest(500)`). The sensitivity sweep (SENSITIVITY_REPORT.md) showed **zero** genes pass the joint-significance test at tv=500. Signal only emerges at tv=2k–5k, with `LOC100249249` as the top mediator ($p = 1.2 \times 10^{-1}$■).

If Figs 49–50 in the existing report show specific mediator names, those names may not be reproducible with the script as written. A reviewer who reruns the code will see zero significant mediators and will file a reproducibility failure.

Remediation

- Re-run `causal_mediation_analysis.py` at tv=2000 (or more). Regenerate

Figs 49–50. Name the stable mediators (cross-referenced to the HIMA sweep in the extended pipeline): `LOC100249249` is the cross-pipeline convergent hit.

- Document in Methods: "Top-variable gene filter at 2,000 selected to

balance multiple-testing burden against signal detection; sensitivity across 500–10,000 reported in Supplementary Fig. Sx."

• Include the full sensitivity table as a supplementary figure.

5. ML biomarker AUC=1.000 claim ■

Issue

The "5 methods achieve AUC=1.000" headline is real for the stated analysis — **but the feature pool includes viral transcripts (`N761_gp1–3`) which are perfectly diagnostic of infection by definition** (you cannot have a viral transcript without the virus). Sensitivity sweep:

Feature pool	AUC range across 5 methods × 4 thresholds
All variable genes (viral + host)	1.000 in every cell
Host-only (viral genes removed)	**0.919 – 1.000** (LASSO at tv=200 is 0.919)

At one sensible threshold (tv=200, LASSO, host-only): **AUC = 0.919**, not 1.000. Reviewers who look at the feature-importance plot in Fig 46 will immediately see N761_gp1/gp3 dominating and ask exactly this question.

Remediation

• Reframe the AUC headline: "Host-transcriptome biomarkers alone achieve AUC = 0.92–1.00 across 5 methods; inclusion of viral transcripts pushes all methods to AUC = 1.000 as expected."

• Report both panels (all-vars and host-only) in Fig 46.

• For the 3-gene host-only panel, report the actual cross-validated AUC on host-only features, not the all-vars AUC with viral genes included.

• Consider leave-one-infection-cohort-out cross-validation as a stricter generalisation test.

6. No qPCR validation of DEGs ■

Issue

Every RNA-seq paper in the target journals validates **at least 5–10 top DEGs** by RT-qPCR on an independent sample aliquot or biological replicate cohort. The repository contains no such validation.

Remediation

• qPCR panel of **at least 10 key genes**:

• 3 top host biomarkers (LOC100854143, LOC100268011, LOC100257913)

• 3 sHSPs (to confirm suppression)

• HSFA2 (master regulator)

• 3 robust temperature-titer mediators (LOC100266309, LOC100232974, LOC100249249)

• Run on a **fresh sample aliquot**, not the same RNA pool used for RNA-seq. Ideally on vines from a subsequent season if possible.

• Expected scope: 6–8 weeks of lab time. If PI won't approve qPCR, a reviewer-safe fallback is to use a published external dataset (e.g. from the meta-analysis targets) to check concordance of the top-50 DEG directions. This is weaker but publishable.

7. GSEA and pathway enrichment framing ■

Issue

• 270 significant gene set × condition pairs at FDR<0.05 with 500 permutations — fine.

• But: what **universe** was used for Fisher ORA in

`pathway_enrichment.py`? All 14,604 filtered genes, or all annotated *V. vinifera* genes? The choice changes p-values substantially.

• No multi-dataset normalisation: GSEA uses t-statistic-ranked genes,

but if the DE method was Welch t (see #1), rankings will differ from DESeq2/edgeR rankings.

Remediation

• State the GSEA ranking metric and universe explicitly in Methods.

• Rerun GSEA on DESeq2 / edgeR-based rankings (follows from #1).

• Confirm that KEGG:Ribosome and GO:translation initiation factor

enrichment (your headline GSEA finding across 6/8 conditions) survives the re-ranking. It probably will — these are robust pathways — but confirm.

8. WGCNA methodology ■

Issue

• 13 modules; soft power β chosen but **scale-free topology R^2 not

reported**. Reviewers commonly ask for `R² > 0.8` at the chosen power.

• Module size distribution looks reasonable but no `mergeCutHeight` /

`deepSplit` / `minModuleSize` reported in methods.

• **The MEs.csv on disk uses sample IDs that don't match counts_matrix.txt**

(`16H1` vs `S01` — documented in CLAUDE.md). This is a legacy artifact. All extended-pipeline downstream analyses (preservation, subtype, etc.) recompute modules from counts. Document this explicitly so a reviewer doesn't get confused trying to cross-reference.

Remediation

• Report scale-free topology R^2 at chosen β in the methods.

• Report all WGCNA parameters (deepSplit, minModuleSize, mergeCutHeight,

networkType).

• Regenerate MEs.csv using the counts_matrix sample IDs so everything

in the repo is self-consistent.

9. Multiple-testing across the whole paper ■

Issue

Each of 6 research questions does its own BH FDR within that question. If the reader is assessing family-wise evidence across 6 questions × up to 8 conditions × 14,604 genes, global FDR is not controlled. A stringent reviewer may ask for a hierarchical or stage-level adjustment.

Remediation

• State clearly: "FDR controlled within each comparison." Usually

sufficient, but flag it.

• For the headline claims (sHSP suppression, mediator panel), report

fold-change + BH q-value together so readers can calibrate.

10. QTL co-localisation claim ■

Issue

"60% biomarker-QTL co-localisation in 1 Mb windows" needs a null. What fraction of 45 random gene sets of matched size would co-localise with 121 QTL intervals by chance? Without a permutation-based null, the 60% could be anywhere from "highly enriched" to "indistinguishable from random" depending on the density of QTL coverage on the Vitis genome.

Remediation

• Permutation test: draw 1,000 random 45-gene sets from the same chromosomes as the biomarkers; compute the null distribution of 1-Mb co-localisations; report empirical p and fold-enrichment.

• Expected: the enrichment will be real but possibly smaller than 60% in effect size. That's fine — just report honestly.

11. Gene annotations are mostly LOC identifiers ■

Issue

Top biomarkers, mediators, and hub TFs are reported as NCBI `LOC` identifiers. This is a pure reproducibility failure at review time — reviewers can't interpret `LOC100249249` without looking it up. For any published paper, every gene named in the Abstract / Figures should have its curated annotation in the main-text caption.

Remediation

• Pull the NCBI gene description for every headline gene (biomarkers, mediators, hub TFs, top pseudotime genes) and add to a supplementary gene-annotation table. `efetch gene -id <GeneID> -format xml` or the NCBI datasets CLI can batch this.

• Replace bare `LOC` references in main-text figures and tables with `LOC100249249 (heat-shock cognate 70-2-like)` style annotations (example — fill with actual gene names after the annotation pass).

12. Causal language in the manuscript ■

Issue

"Causal mediation" and similar terms appear in the pipeline and docs. Observational mediation with n ≈ 60 vines per season is **causal under specific assumptions** (no unmeasured confounders, correct DAG, no reverse causation). Plant-virology reviewers are mixed on whether to accept "causal" without an intervention.

Remediation

• Use "statistical mediation" or "objective-inference mediation" in the Methods and Results unless and until the VIGS experiment is run.

• In the Discussion, frame the mediator panel as "high-priority candidates for functional validation" rather than "confirmed causal mediators."

• If aiming at PNAS, the VIGS experiment upgrades the language to "causal" legitimately.

13. Missing external validation ■

Issue

Everything is internal to one 327-sample cohort from Merlot at a single vineyard. Even within the dataset, 5-fold CV (Fig 46) uses random sample assignment, which is not a generalisation test — samples from the same vine or the same year end up on both sides of the split.

Remediation

• **Leave-one-cohort-out (LOCO) CV**: 7 infection cohorts × 2 seasons = up to 14 held-out groups. Train on 13, test on the 14th. Report AUC across folds. This is the actual "generalisation" test. If AUC drops substantially, the paper's diagnostic claim softens accordingly.

• **External validation with published Vitis stress datasets** — this is what rules 11–12 (Tier 1 meta-analysis, scaffolded) will deliver. Priority: run Tier 1 before submission.

14. Figure-level issues a reviewer will flag ■

Quick-scan issues likely to come up in review:

• **Color choice**: verify color-blind safe palettes for all heatmaps (viridis / RdBu_r is fine; tab10 / jet is not).

• **Axis ranges inconsistent** across volcano plots in Figs 20, 22, 26 — match them or state why they differ.

• **Gene names illegible** at submission DPI. Recompute at final journal size before submission.

• **Figure 14 master pathway heatmap** is information-dense; consider splitting into two or moving to a supplement.

• **Network figure (Fig 53–55)** — reviewers often want the full edge list as a supplementary TSV. Available? Confirm.

15. The sensitivity sweep itself is a strength ■

Not an issue, but worth calling out: the fact that every threshold bottleneck has now been swept is **above-average rigour** for a plant RNA-seq paper. Include the sensitivity report as a supplementary file and cite it in the Methods. This converts a weakness (threshold sensitivity of specific gene claims) into a demonstration of methodological care.

Summary — pre-submission checklist

Order of priority:

1. ■ Refit DE with DESeq2 or limma-voom; regenerate all downstream gene lists (§1).
2. ■ Reproduce QC + alignment stats; produce MultiQC; deposit to GEO (§3).
3. ■ Rerun `causal_mediation_analysis.py` at tv ≥ 2,000; regenerate Figs 49–50 (§4).
4. ■ Reframe the AUC=1.000 claim with the host-only caveat (§5).
5. ■ Perform qPCR validation of ≥ 10 key genes (§6).
6. ■ Annotate every gene named in the paper by its NCBI product (§11).
7. ■ LOCO cross-validation for the biomarker panel (§13).
8. ■ QTL co-localisation permutation test (§10).
9. ■ Add repeated-measures term to the DE model (§2).
10. ■ Soften "causal" language to "statistical mediation" until VIGS is done (§12).
11. ■ Report WGCNA parameters and scale-free R^2 (§8).
12. ■ Tier-1 meta-analysis (rules 11–12) as external validation (§13).
13. ■ Figure typography / palette pass (§14).

Time estimate to address everything above: 4–6 weeks focused work on the compute side, plus 6–8 weeks of lab time for qPCR (parallelisable with analysis work).

After these fixes, the paper is Nature Plants-competitive as-is. With the HSFA2 VIGS experiment on top, it's PNAS-competitive.

Appendix: journal-specific red lines

Nature Plants. Expects state-of-the-art statistical methods (DESeq2 / limma-voom mandatory in practice), qPCR validation, transparent data availability, clean Methods. VIGS not strictly required if the causal inference is rigorous.

Plant Cell. Similar but more tolerant of a purely bioinformatic paper with solid qPCR. A functional experiment strengthens the case but is not mandatory for landing.

New Phytologist / Plant Journal. More permissive on methods; qPCR still expected. The current pipeline, after fixes 1–6 above, fits well here.

PNAS. Effectively mandatory functional validation (VIGS, CRISPR, heterologous expression, or biochemical assay) for plant-virology papers. Without it, the paper will struggle at editorial triage.

BMC Genomics / MPMI. Forgiving on methodology; the paper is submission-ready for these journals after fixes 1, 3, 4, 5 (the four red items).

This critical review was generated by comparing the existing analysis pipeline against the methodological conventions of published plant RNA-seq studies in the target journals over 2020–2025. It is intentionally pessimistic (a reviewer's first pass) and should be treated as a pre-submission stress test, not a final judgement.